

Midterm Report

SOC-Introduction to Machine Learning

Week1:

Learnt Python Basics Along with a general Introduction to Machine Learning.

Week2:

Learnt

- Numpy
- Pandas
- Matplotlib
- Seaborn

Week3:

Learnt

- Linear Regression and Model Evaluation
- R^2 Score and Model Evaluation
- Ridge and Lasso Regression
- Linear Regression implementation using Scikit-Learn
- Train-Test Split demonstration
- Visualization of regression results
- Basic understanding of Gradient Descent

Week 4:

Learnt

- Odds and Log-Odds.
- The intuition behind Logistic Regression.
- Coefficients and Maximum Likelihood Estimation.
- Prediction probabilities and decision boundaries.
- Binary and multiclass classification.

Each Week had its own assignments which were completed on time.